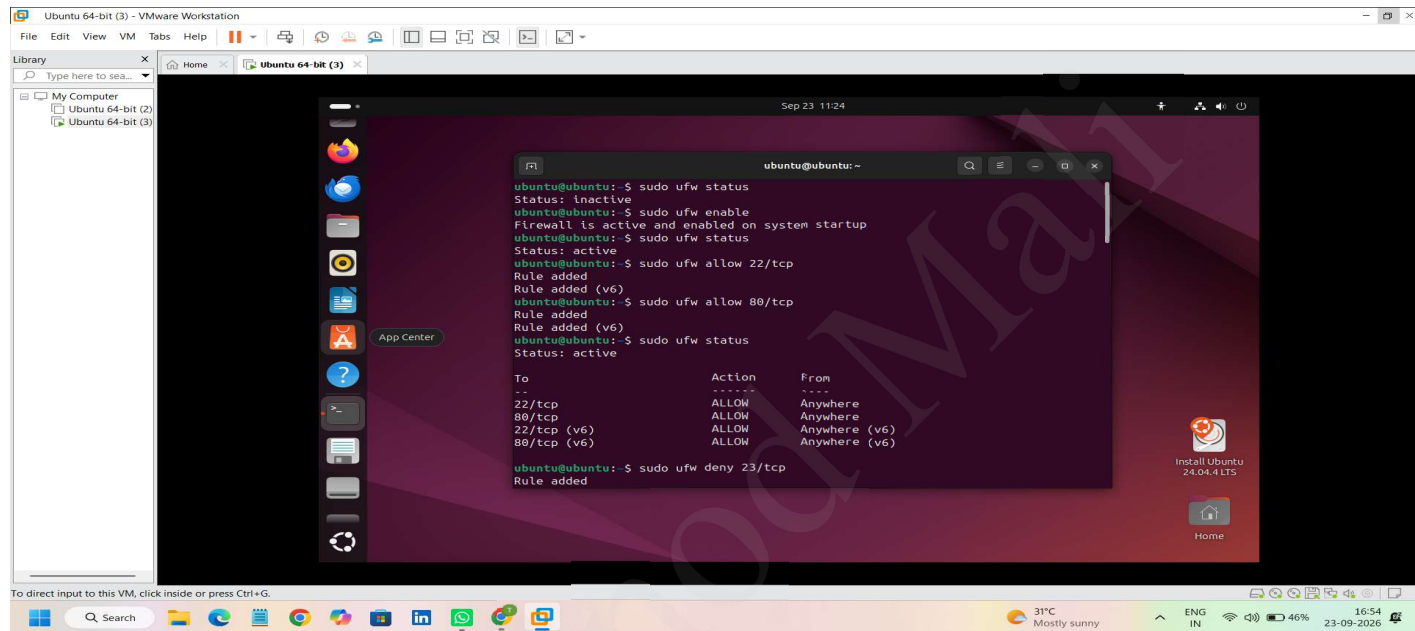


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Aim:- Configure and test firewall rules to control network traffic, filter packets based on specified criteria, and protect network resources from unauthorized access.

Step 1: Open Ubuntu Terminal

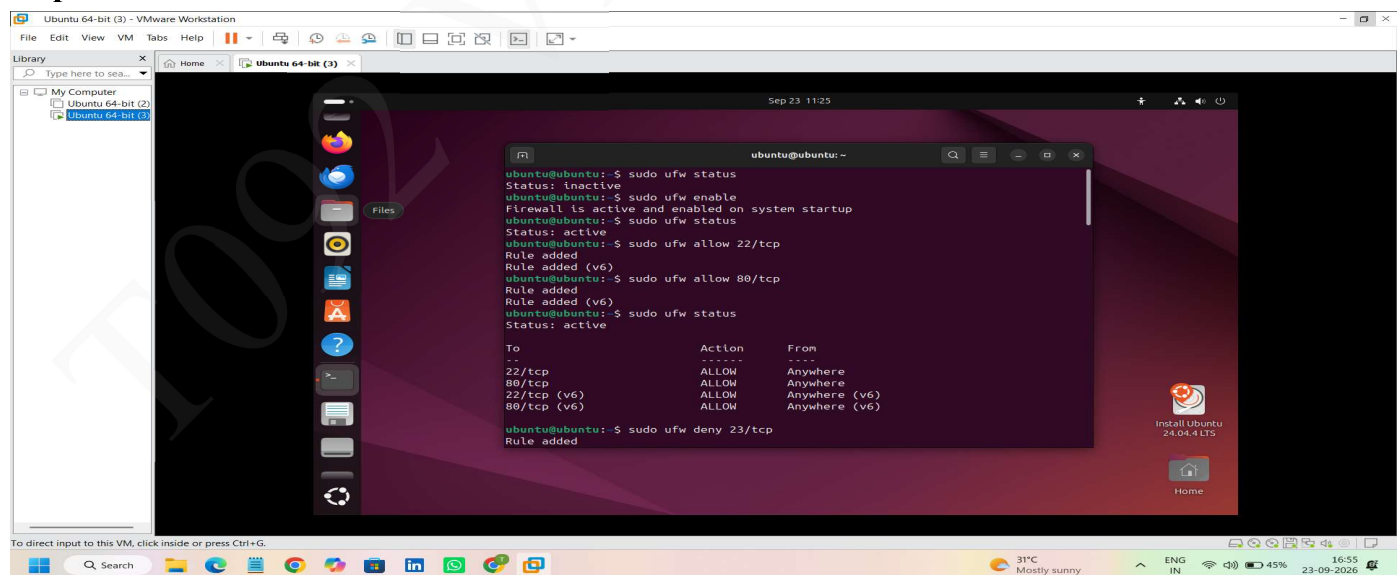


```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo ufw status
Status: active
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
```

Step 2: Enable the Firewall



```
ubuntu@ubuntu:~$ sudo ufw status
Status: inactive
ubuntu@ubuntu:~$ sudo ufw enable
Firewall is active and enabled on system startup
ubuntu@ubuntu:~$ sudo ufw status
Status: active
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

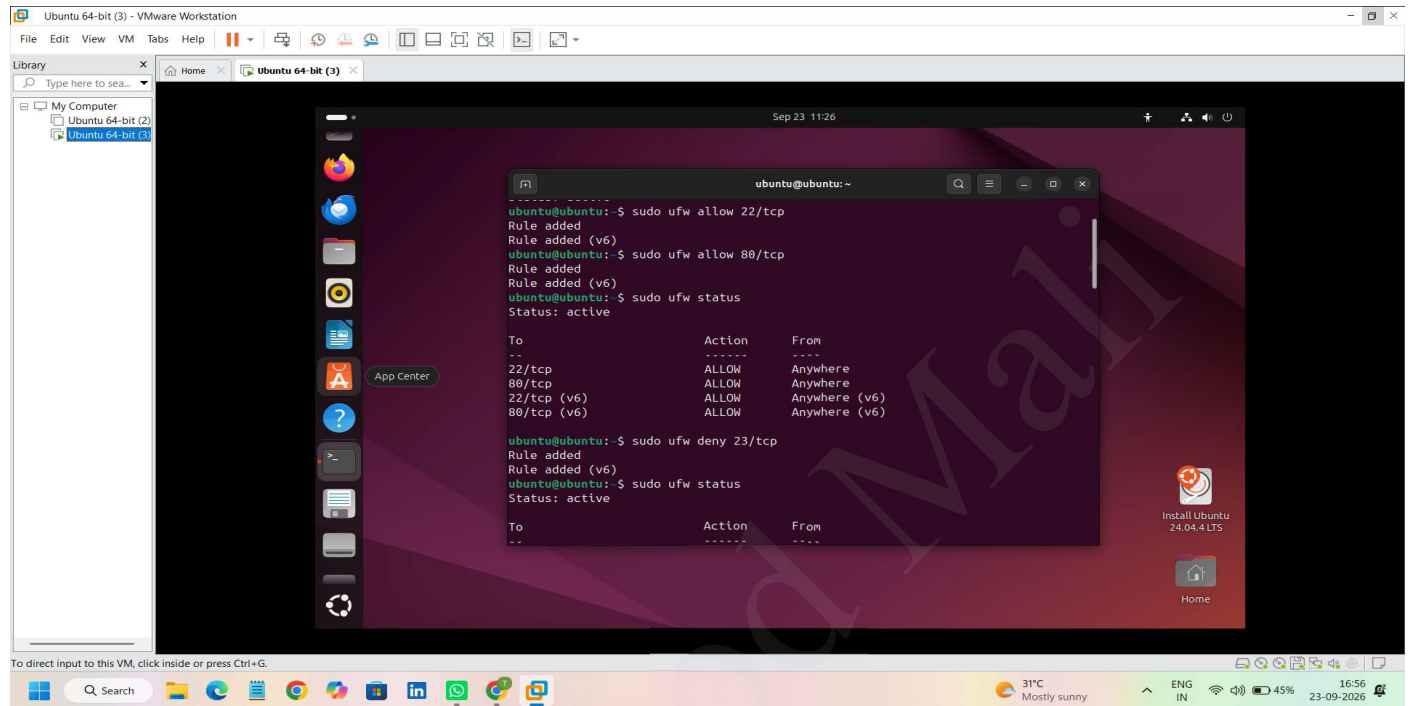
To Action From
--
22/tcp ALLOW Anywhere
80/tcp ALLOW Anywhere
22/tcp (v6) ALLOW Anywhere (v6)
80/tcp (v6) ALLOW Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
```

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Step 3: Allow SSH Traffic



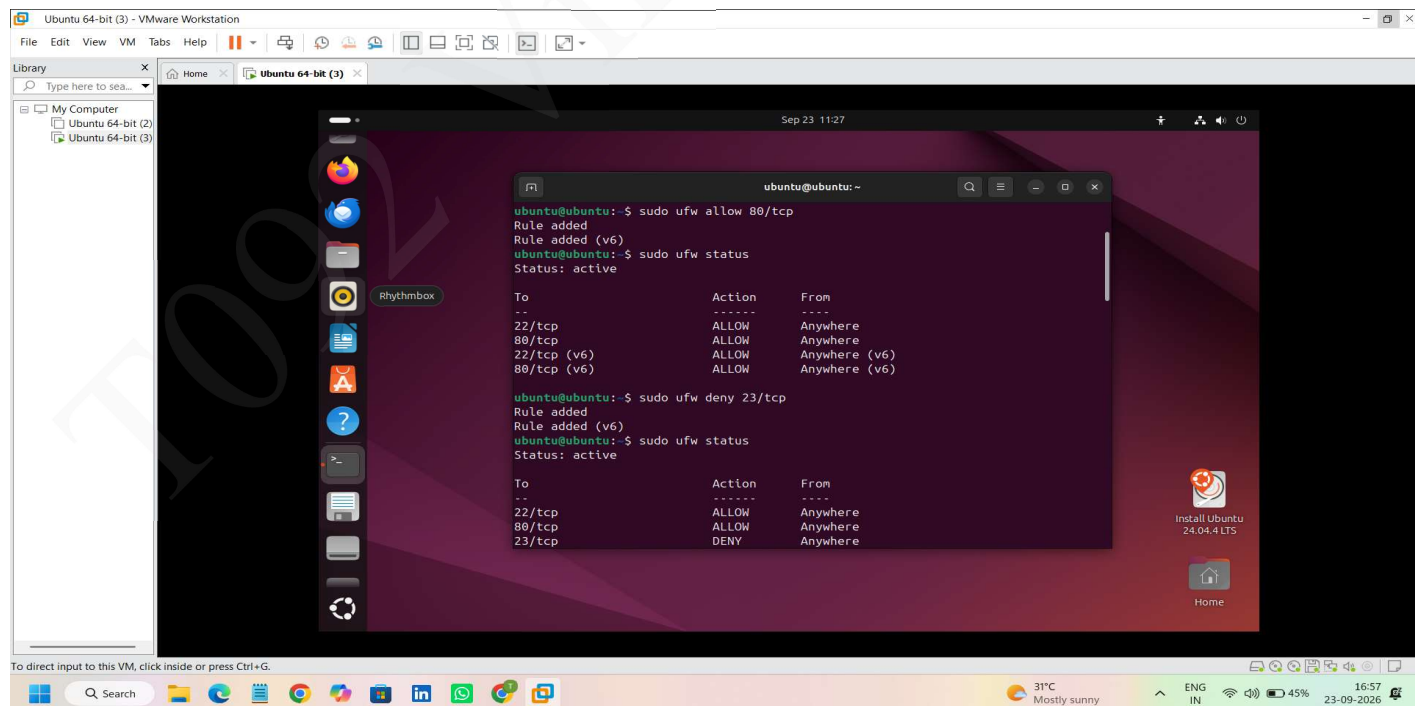
```
ubuntu@ubuntu:~$ sudo ufw allow 22/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To                Action      From
--                -
22/tcp            ALLOW      Anywhere
80/tcp            ALLOW      Anywhere
22/tcp (v6)       ALLOW      Anywhere (v6)
80/tcp (v6)       ALLOW      Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To                Action      From
--                -
22/tcp            ALLOW      Anywhere
80/tcp            ALLOW      Anywhere
23/tcp            DENY       Anywhere
```

Step 4:- Step 4: Allow HTTP Traffic



```
ubuntu@ubuntu:~$ sudo ufw allow 80/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To                Action      From
--                -
22/tcp            ALLOW      Anywhere
80/tcp            ALLOW      Anywhere
22/tcp (v6)       ALLOW      Anywhere (v6)
80/tcp (v6)       ALLOW      Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw deny 23/tcp
Rule added
Rule added (v6)
ubuntu@ubuntu:~$ sudo ufw status
Status: active

To                Action      From
--                -
22/tcp            ALLOW      Anywhere
80/tcp            ALLOW      Anywhere
23/tcp            DENY       Anywhere
```

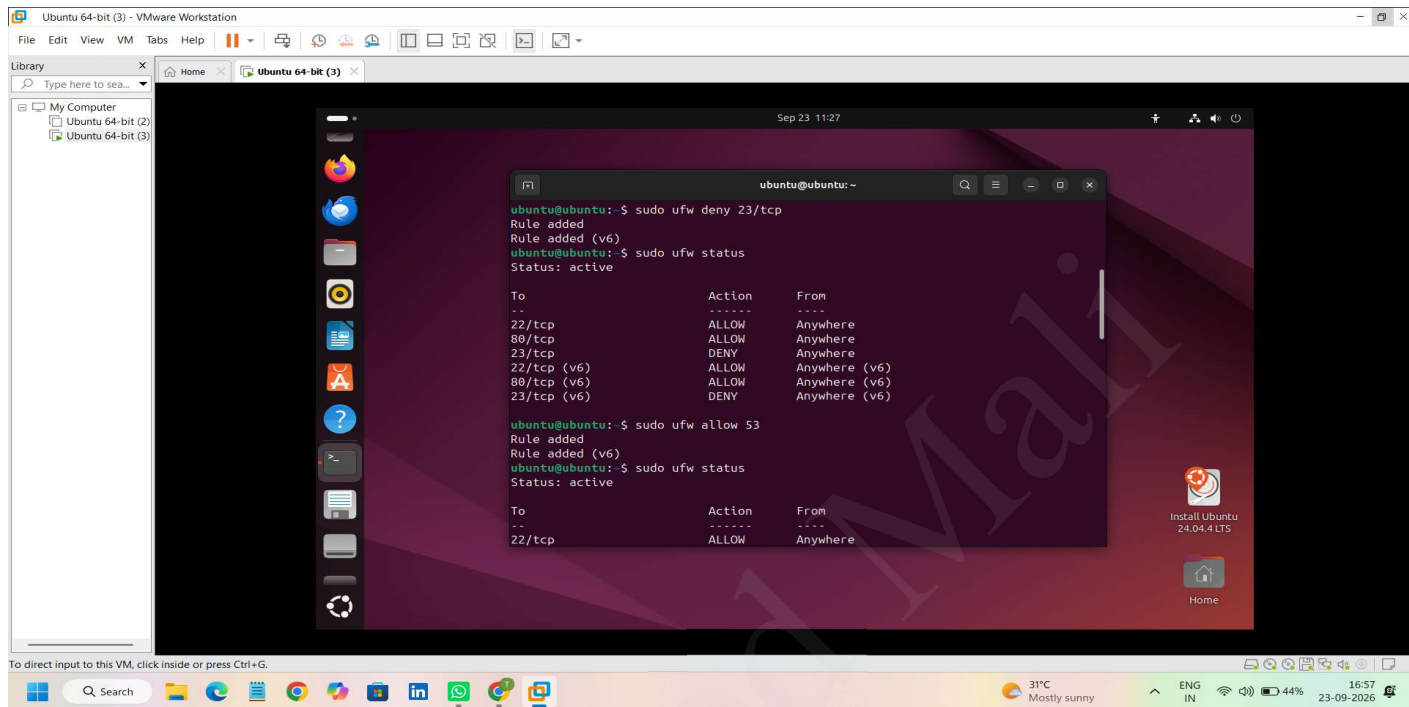
Step 5: Block a Port

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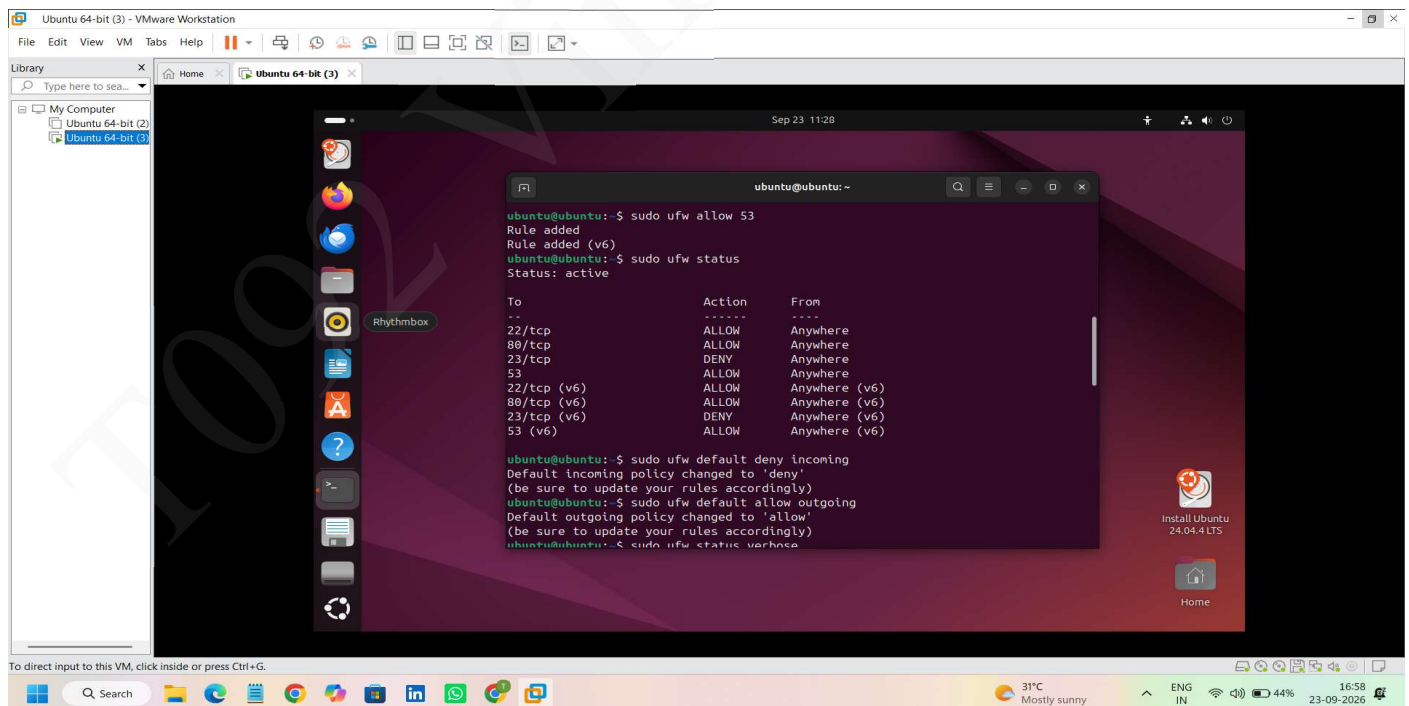
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Step 6: Allow DNS Traffic



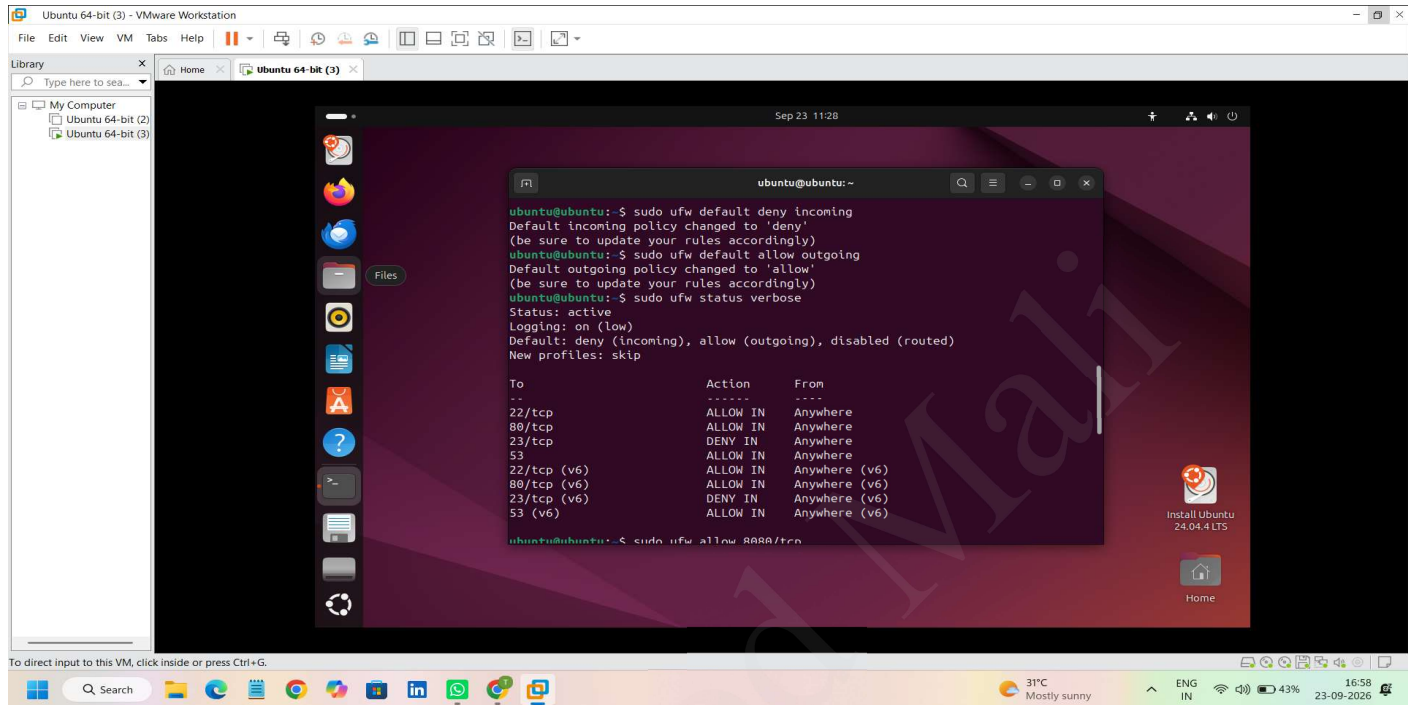
Step 7: Set Default Firewall Policies

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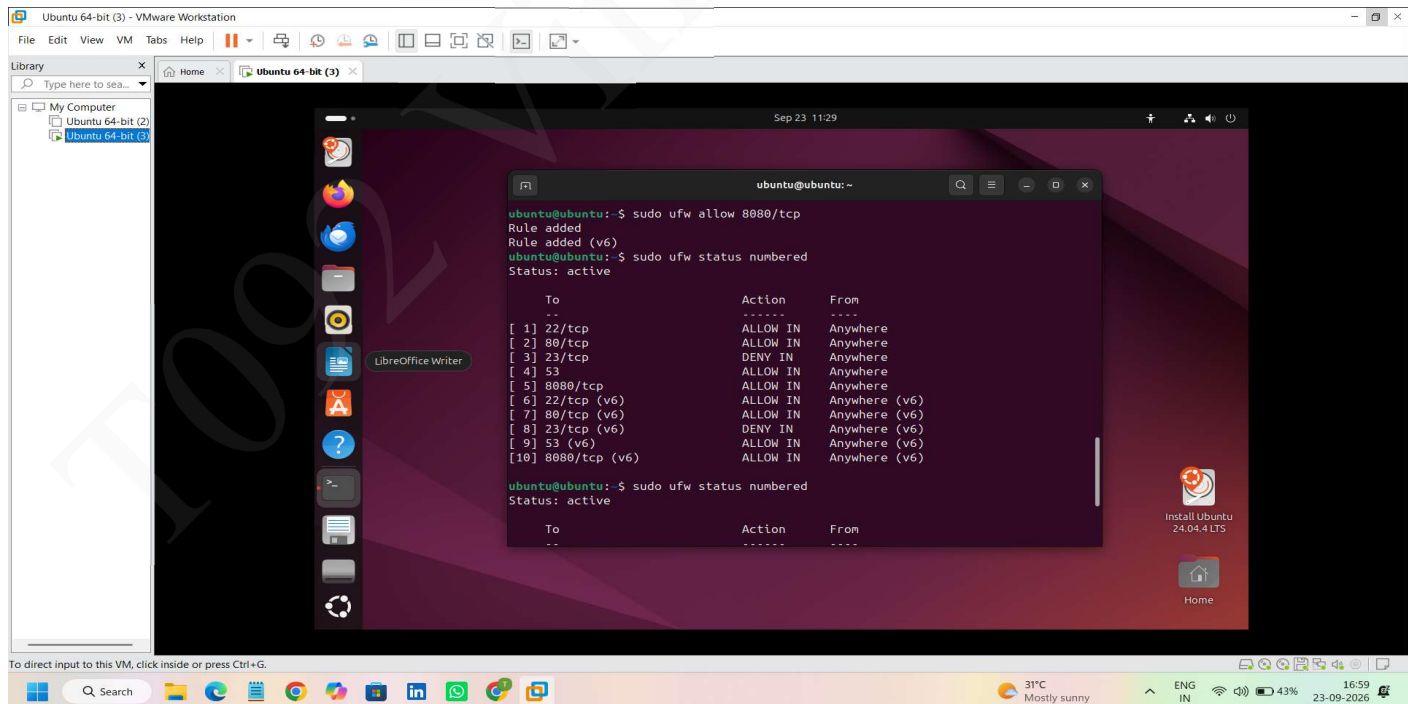
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Step 8: Create a Rule for a Specific Port



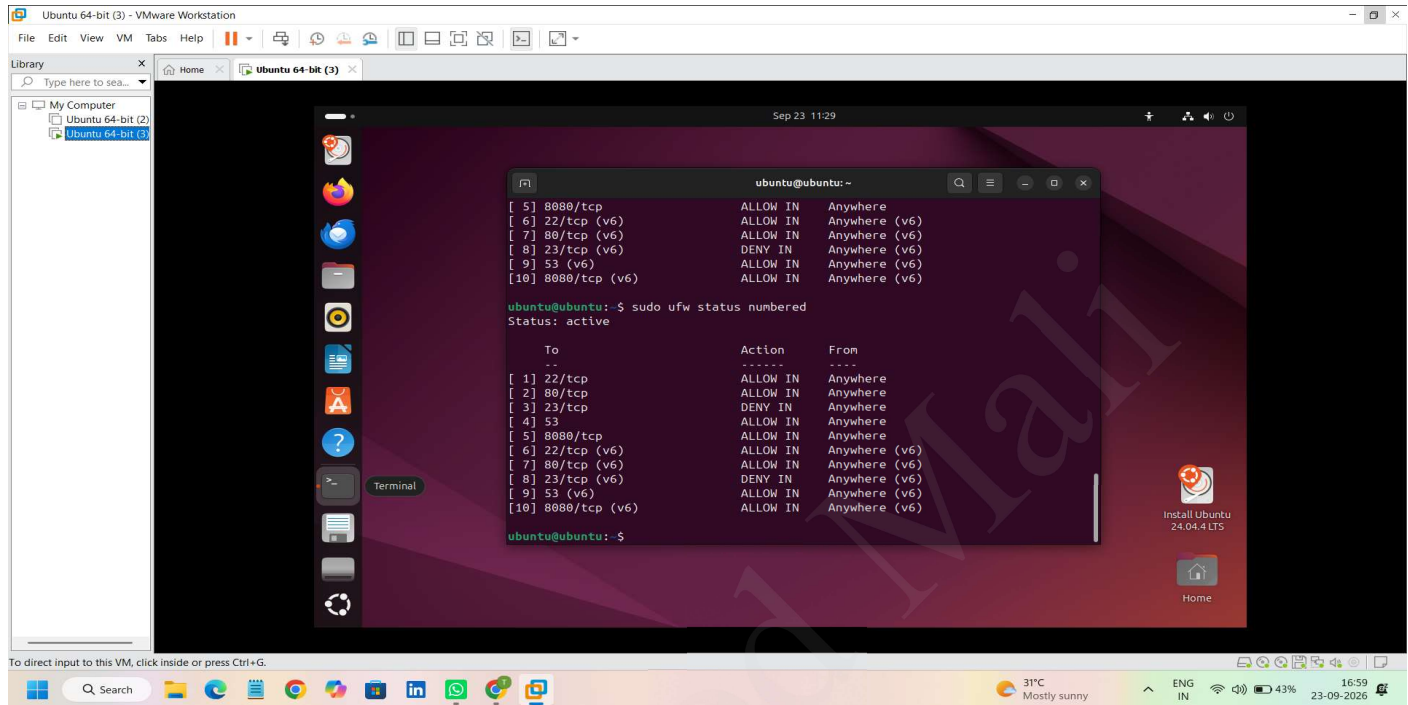
Step 9: Test the Firewall

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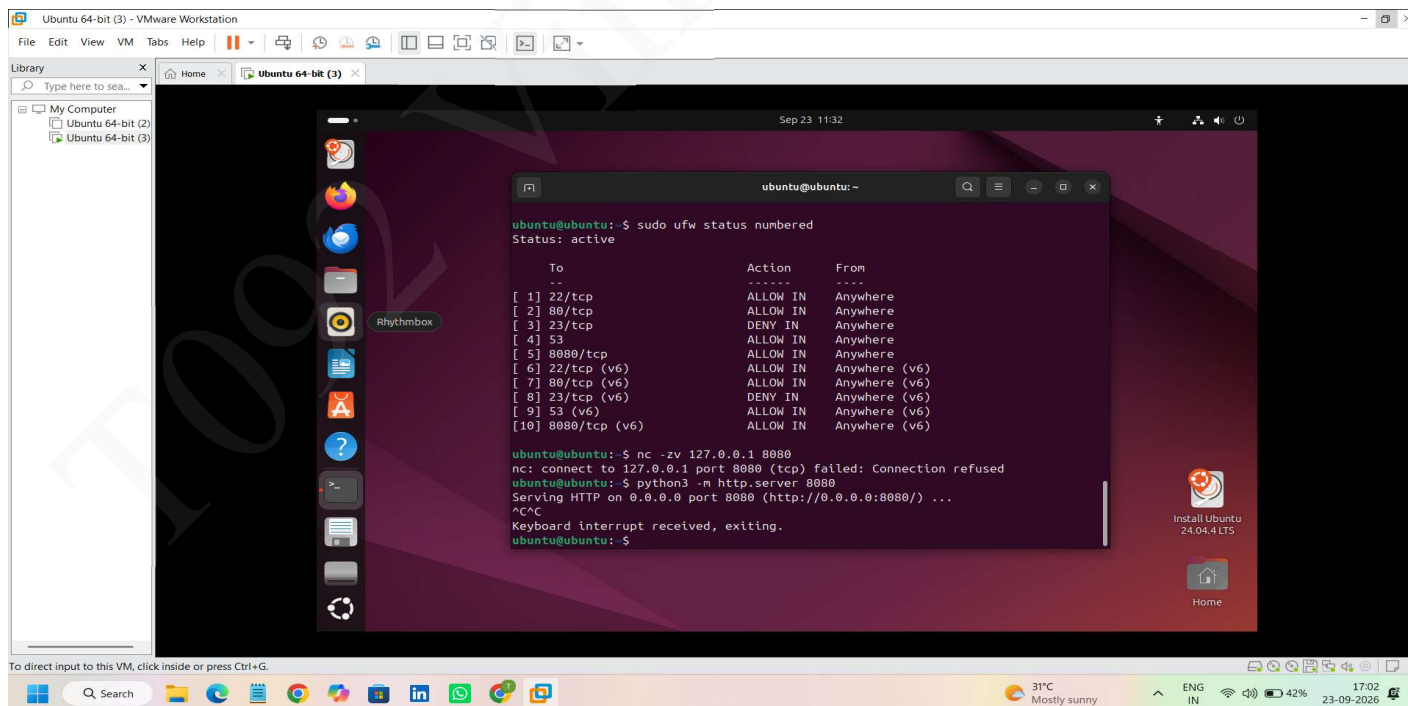
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```
ubuntu@ubuntu:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 23/tcp DENY IN Anywhere
[ 4] 53 ALLOW IN Anywhere
[ 5] 8080/tcp ALLOW IN Anywhere
[ 6] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 7] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 8] 23/tcp (v6) DENY IN Anywhere (v6)
[ 9] 53 (v6) ALLOW IN Anywhere (v6)
[10] 8080/tcp (v6) ALLOW IN Anywhere (v6)
```



```
ubuntu@ubuntu:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 23/tcp DENY IN Anywhere
[ 4] 53 ALLOW IN Anywhere
[ 5] 8080/tcp ALLOW IN Anywhere
[ 6] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 7] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 8] 23/tcp (v6) DENY IN Anywhere (v6)
[ 9] 53 (v6) ALLOW IN Anywhere (v6)
[10] 8080/tcp (v6) ALLOW IN Anywhere (v6)

ubuntu@ubuntu:~$ nc -zv 127.0.0.1 8080
nc: connect to 127.0.0.1 port 8080 (tcp) failed: Connection refused
ubuntu@ubuntu:~$ python3 -m http.server 8080
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...
^C
Keyboard interrupt received, exiting.
ubuntu@ubuntu:~$
```

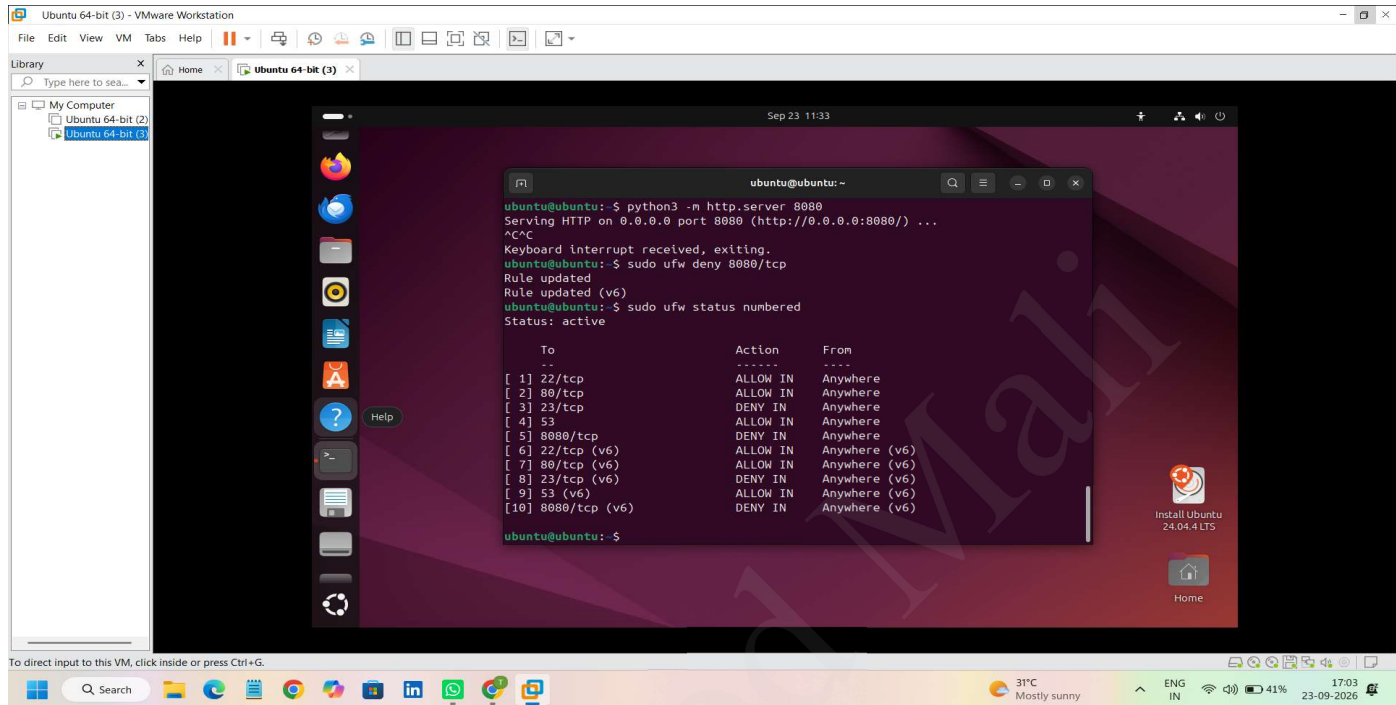
Step 10: Test Blocking

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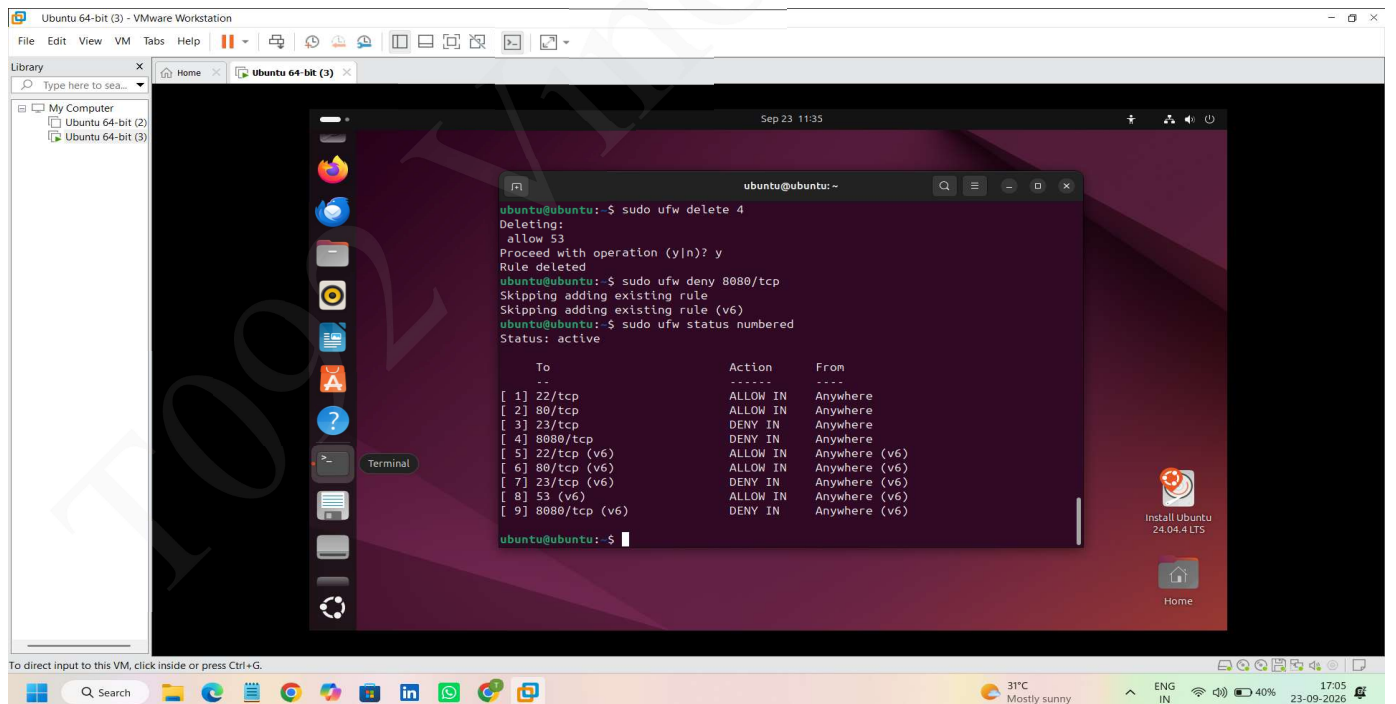
Practical no 10



```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ python3 -m http.server 8080  
Serving HTTP on 0.0.0.0 port 8080 (http://0.0.0.0:8080/) ...  
^C^C  
Keyboard interrupt received, exiting.  
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp  
Rule updated  
Rule updated (v6)  
ubuntu@ubuntu:~$ sudo ufw status numbered  
Status: active  


| To                 | Action   | From          |
|--------------------|----------|---------------|
| [ 1] 22/tcp        | ALLOW IN | Anywhere      |
| [ 2] 80/tcp        | ALLOW IN | Anywhere      |
| [ 3] 23/tcp        | DENY IN  | Anywhere      |
| [ 4] 53            | ALLOW IN | Anywhere      |
| [ 5] 8080/tcp      | DENY IN  | Anywhere      |
| [ 6] 22/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 7] 80/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 8] 23/tcp (v6)   | DENY IN  | Anywhere (v6) |
| [ 9] 53 (v6)       | ALLOW IN | Anywhere (v6) |
| [10] 8080/tcp (v6) | DENY IN  | Anywhere (v6) |

  
ubuntu@ubuntu:~$
```



```
ubuntu@ubuntu: ~  
ubuntu@ubuntu:~$ sudo ufw delete 4  
Deleting:  
allow 53  
Proceed with operation (y/n)? y  
Rule deleted  
ubuntu@ubuntu:~$ sudo ufw deny 8080/tcp  
Skipping adding existing rule  
Skipping adding existing rule (v6)  
ubuntu@ubuntu:~$ sudo ufw status numbered  
Status: active  


| To                 | Action   | From          |
|--------------------|----------|---------------|
| [ 1] 22/tcp        | ALLOW IN | Anywhere      |
| [ 2] 80/tcp        | ALLOW IN | Anywhere      |
| [ 3] 23/tcp        | DENY IN  | Anywhere      |
| [ 4] 8080/tcp      | DENY IN  | Anywhere      |
| [ 5] 22/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 6] 80/tcp (v6)   | ALLOW IN | Anywhere (v6) |
| [ 7] 23/tcp (v6)   | DENY IN  | Anywhere (v6) |
| [ 8] 53 (v6)       | ALLOW IN | Anywhere (v6) |
| [ 9] 8080/tcp (v6) | DENY IN  | Anywhere (v6) |

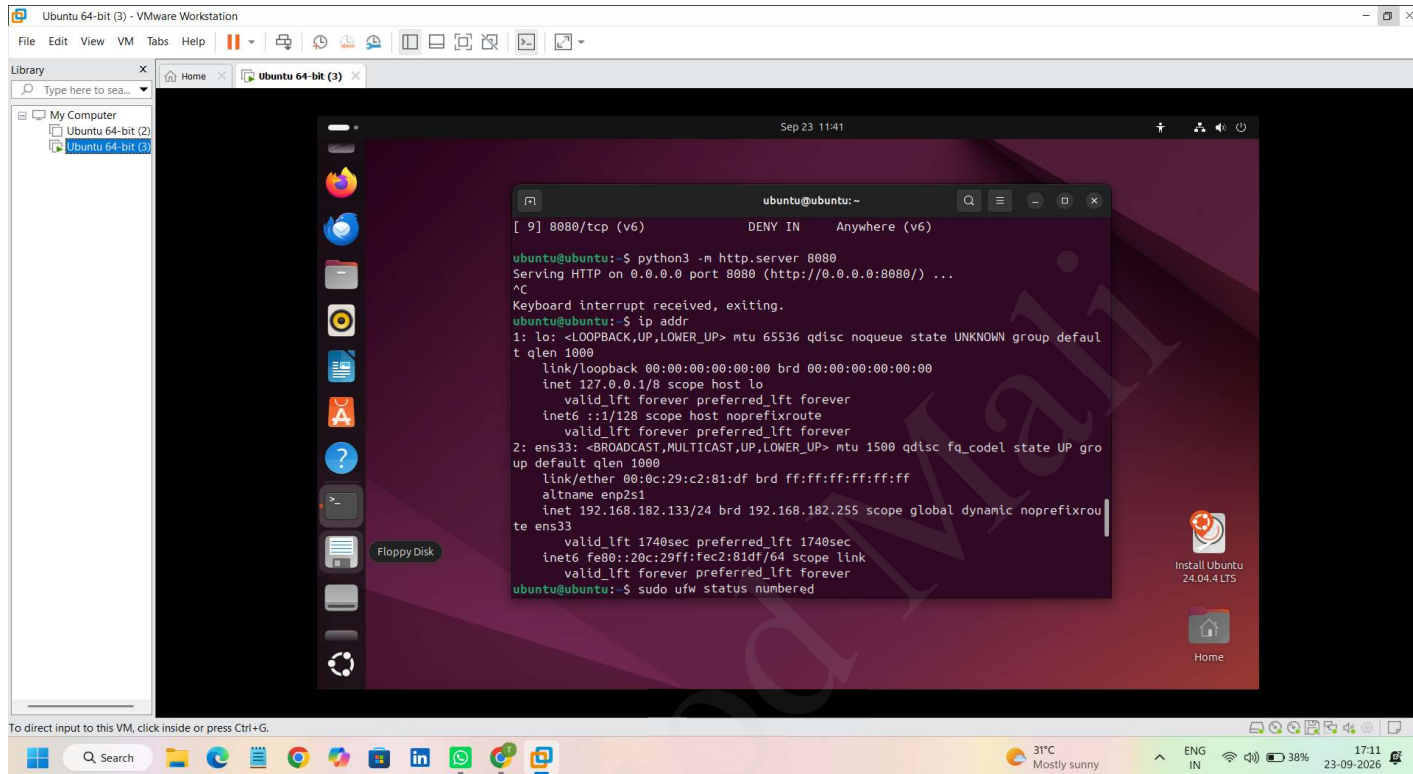
  
ubuntu@ubuntu:~$
```

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Ubuntu 64-bit (3) - VMware Workstation

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- Ubuntu 64-bit (2)
- Ubuntu 64-bit (3)

Home

Install Ubuntu 24.04.4 LTS

Home

```
ubuntu@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:c2:81:df brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    inet 192.168.182.133/24 brd 192.168.182.255 scope global dynamic noprefixroute
        valid_lft 1740sec preferred_lft 1740sec
    inet6 fe80::20c:29ff:fec2:81df/64 scope link
        valid_lft forever preferred_lft forever
ubuntu@ubuntu:~$ sudo ufw status numbered
```

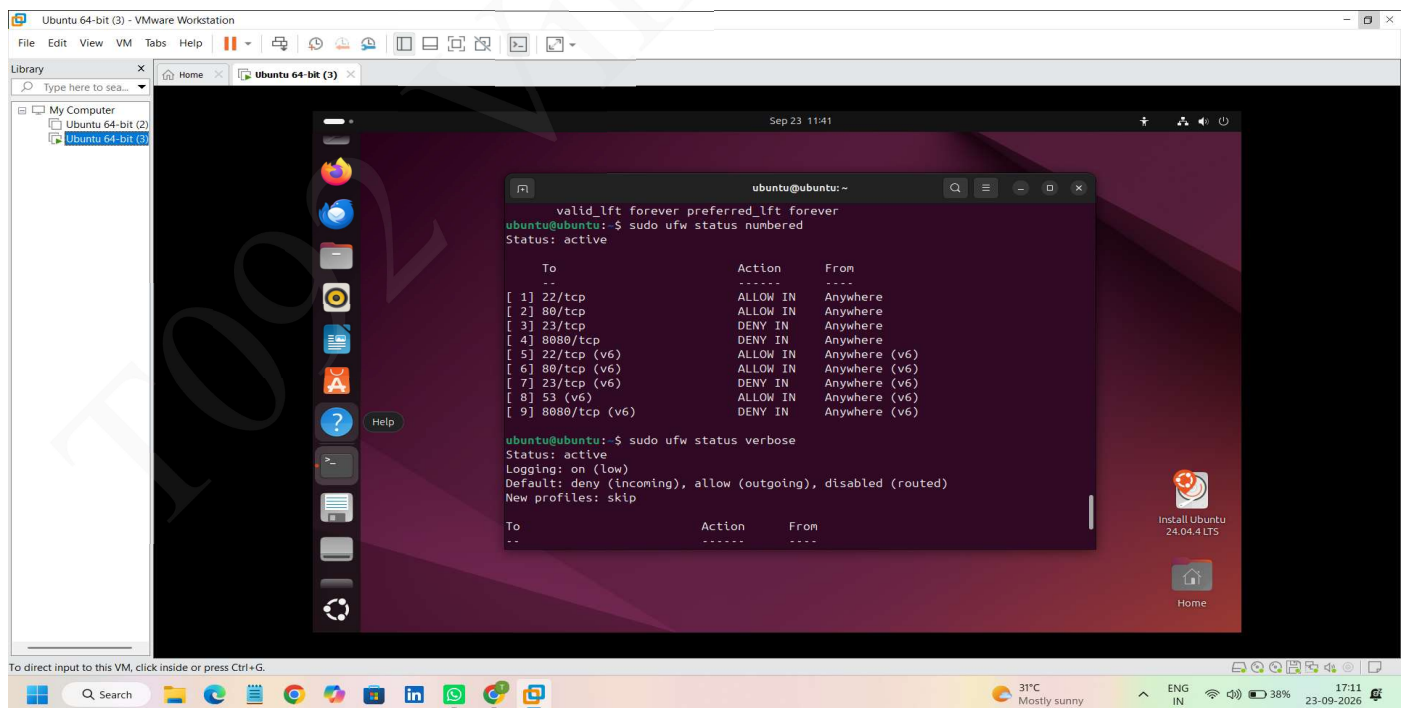
To direct input to this VM, click inside or press Ctrl+G.

Search

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ENG IN

17:11 23-09-2026



Ubuntu 64-bit (3) - VMware Workstation

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Library

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My Computer

- Ubuntu 64-bit (2)
- Ubuntu 64-bit (3)

Home

Install Ubuntu 24.04.4 LTS

Home

```
ubuntu@ubuntu:~$ sudo ufw status numbered
Status: active

To Action From
--
[ 1] 22/tcp ALLOW IN Anywhere
[ 2] 80/tcp ALLOW IN Anywhere
[ 3] 23/tcp DENY IN Anywhere
[ 4] 8080/tcp DENY IN Anywhere
[ 5] 22/tcp (v6) ALLOW IN Anywhere (v6)
[ 6] 80/tcp (v6) ALLOW IN Anywhere (v6)
[ 7] 23/tcp (v6) DENY IN Anywhere (v6)
[ 8] 53 (v6) ALLOW IN Anywhere (v6)
[ 9] 8080/tcp (v6) DENY IN Anywhere (v6)

ubuntu@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
```

To direct input to this VM, click inside or press Ctrl+G.

Search

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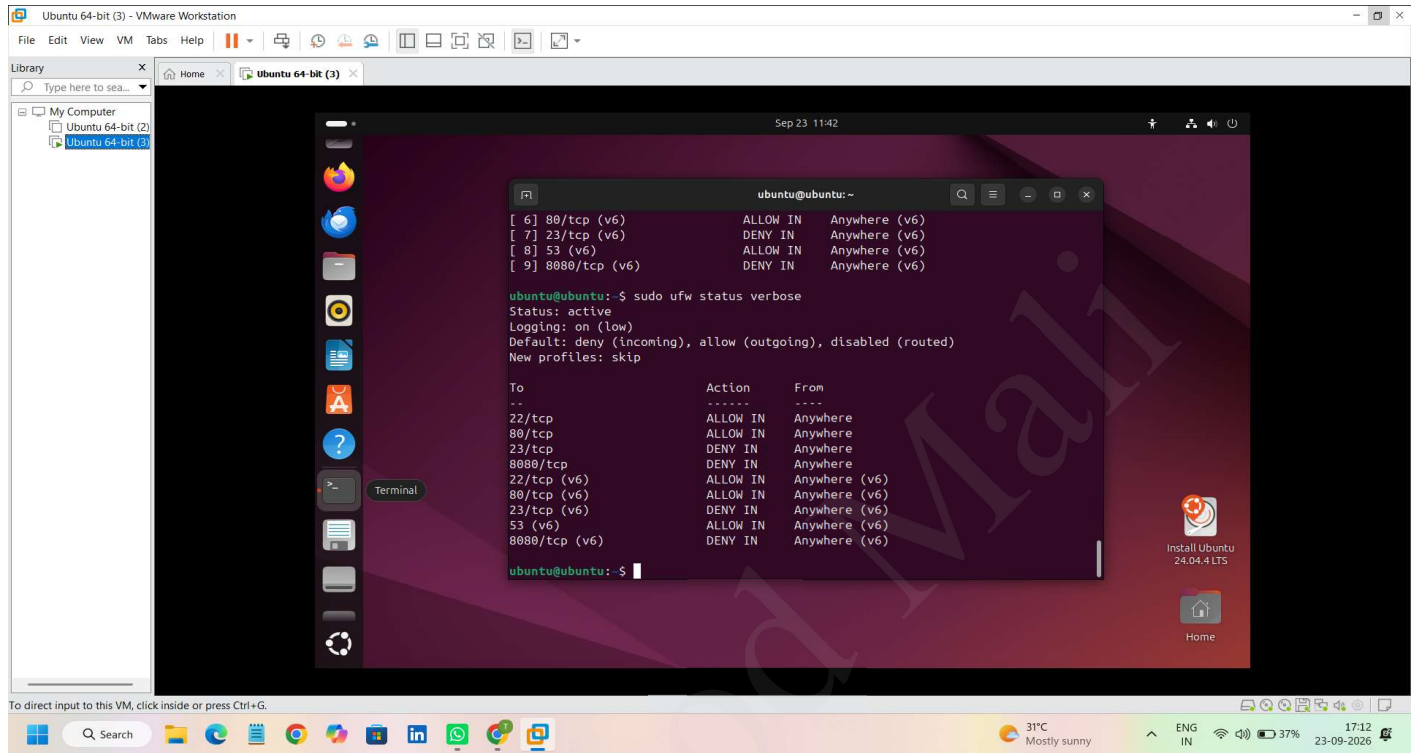
ENG IN

17:11 23-09-2026

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The screenshot shows a VMware Workstation window titled "Ubuntu 64-bit (3) - VMware Workstation". Inside the VM, the Ubuntu desktop is visible with a terminal window open. The terminal output is as follows:

```
ubuntu@ubuntu:~$ sudo ufw status verbose
Status: active
Logging: on (low)
Default: deny (incoming), allow (outgoing), disabled (routed)
New profiles: skip

To Action From
--
22/tcp ALLOW IN Anywhere
80/tcp ALLOW IN Anywhere
23/tcp DENY IN Anywhere
8080/tcp DENY IN Anywhere
22/tcp (v6) ALLOW IN Anywhere (v6)
80/tcp (v6) ALLOW IN Anywhere (v6)
23/tcp (v6) DENY IN Anywhere (v6)
53 (v6) ALLOW IN Anywhere (v6)
8080/tcp (v6) DENY IN Anywhere (v6)
```

The terminal prompt is `ubuntu@ubuntu:~$`. The desktop background is a dark purple Ubuntu logo. The taskbar on the left shows various application icons. The bottom of the screen shows the VMware Workstation interface with a Windows taskbar at the very bottom.

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